

NIRAV SURABHI

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EDUCATION

Purdue University

Bachelor of Science, Mechanical Engineering

West Lafayette, IN

Expected Graduation: May 2029

Relevant Coursework: ENGR 133, MA 265 Linear Algebra, PHYS 172 Modern Mechanics | Activities: Purdue Formula SAE (Fall 2026, design-onboarding sprint in CAD)

Independence High School

High School Diploma, GPA: 3.8/4.00

Loudoun County, VA

June 2026

Relevant Coursework: AP Physics C (Mechanics and E&M), Multivariable Calculus, AP Computer Science A

Honors: Mathematics Scholar, Central Virginia Governor's School for Science & Technology; Presidential Volunteer Service Award, Silver

ENGINEERING AND RESEARCH EXPERIENCE

Passive Vortex Propulsion in Air: A Proof of Concept Demonstration — IEEE EngiTek 2025 — *First Author* 2024 – Dec 2025
& *Principal Investigator*

- Engineered and published the first experimental demonstration of passive vortex propulsion in air: 3.1 N mean net thrust over 20 trials ($p < 0.001$) from a fin placed in a bluff-body wake.
- Designed and fabricated the test equipment from scratch: a custom wind tunnel, bluff-body and fin mounts, and a rig for repeatable wake spacing.
- Built the data-acquisition system: soldered load cells to an Arduino, calibrated the sensor array, and validated readings against analytical models.
- Compared 3D-printed flat and NACA 0024/0012/0009 fins with Fusion 360 CFD and ANOVA/Dunn's tests in Python; NACA 0012 won.
- Won the AIAA Aerospace Award at the Regional Science & Engineering Fair, with an invitation to collaborate at NASA Goddard.

I3 Simulations — CAD Intern

Jun 2025 – May 2026

- Designed 3D CAD models and prototyped haptic-feedback systems with cross-functional developers for medical-training simulators.
- Conducted prototype verification testing for usability, reliability, and conformance to design requirements.

FIRST Tech Challenge Robotics, Absolute Zero #12096 — Team Captain & Lead Engineer

Sep 2022 – Jun 2026

- Directed end-to-end hardware development over four seasons: Fusion 360 CAD of every mechanism, structural fabrication, and electrical wiring; advanced to the Chesapeake regional championship.
- Performed component-level troubleshooting and repair, including precision soldering and refurbishment of failing servo motors.
- Integrated hardware with Java control software: PID loops, encoder and gyroscope sensor fusion, and odometry, verified for timing and signal accuracy.

The Bike Lane — Bike Mechanic (Intern, promoted to Part-Time Staff)

Jun 2021 – Aug 2025

- Assembled bicycles from the frame up for the sales floor and ran the final safety inspection on every build before customer delivery.
- Performed root-cause investigations on nonconforming units and executed precision repairs on drivetrain, brake, and shifting systems.

TECHNICAL PROJECTS

Motor-Based Resistance Trainer — Designer & Builder

Mar 2025

- Built three prototypes (eddy-current, chain-driven, geared) of a portable, battery-powered strength trainer; a 188:1 geared motor under Arduino PWM control gave a linear, repeatable force curve ($R^2 = 0.993$) for \$215.

Science Olympiad, Independence High School — Team Captain & Lead Hardware Engineer

- Engineered a 3D-printed, Arduino-driven electric vehicle, tuning its PID loop through iterative testing for accurate distance runs; built and calibrated an Arduino electrode sensor that measures water salinity.

SpareMe — spareme.com — Founder & Engineer

Jul 2026 – Present

- Shipped an App Store iPhone app and Node/TypeScript backend that delegate phone calls to an AI agent (Twilio real-time speech, OpenAI), with fail-closed kill switches and rate limits.

LEADERSHIP AND INVOLVEMENT

Ember Learning (formerly DeAP Learning Labs) — AI Engineer & Software Intern

Feb 2021 – 2022

- Engineered LLM prompts and automated rubric-scoring tests for AP-tutor chatbots; team won a Microsoft and OpenAI grant.

RELEVANT SKILLS

Mechanical Design: Fusion 360 (CAD, FEA, CFD), design for additive (FDM) and subtractive manufacturing, fabrication, rapid prototyping

Electromechanical: assembly, soldering, wiring, Arduino, sensor calibration (load cells, electrodes, encoders, gyroscopes), motor repair

Analysis & Testing: test-equipment design, verification testing, PID control, fluid dynamics, ANOVA statistics, root-cause analysis

Software: Python, Java, C++ (Arduino), Swift, TypeScript, Git, technical documentation